

Mahmoud Sajjadi

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Research Interests

Control, Power & Machine Learning Engineer (Ph.D. | CDCE) — energy infrastructure, electrical distribution, grid integration, and DCIM. Manages technical teams and troubleshoots data center architecture, HVAC/cooling, networking, UPS, and diesel generators. Bridges control frameworks, mathematical optimization, predictive modeling, scalable AI, and PMU data analytics to cut operational costs, optimize equipment efficiency, ensure EHS compliance, and prevent outages.

Data Center, Power, Control, & Energy Systems Experience

Computer Science and Engineering Department, University of Nevada, Reno, NV

Graduate Research Assistant

2022 – Present

- Engineered federated ML classifiers (XGBoost, MLP) to detect current, voltage, and frequency faults in electrical systems, achieving 91% accuracy utilizing only 80% of local decentralized data (Presented in PESGM conference).
- Developed Distributed ML and GenAI strategies to optimize EV ecosystems (smart charging, V2G load balancing), identifying frameworks that boost battery State of Health (SOH) estimation accuracy by 44.5% (1.16% MAE) and achieve 100% anomaly detection in CAN bus networks.
- Designed a privacy-preserving federated framework with additive homomorphic encryption to classify 3,877 power grid events across 23 PMUs, sustaining 88–90% analytics accuracy on highly non-IID data distributions.

Prochista (Sematec, MCI), Tehran, Iran

Data Center Designer and Manager, Lecturer, Researcher

2019 – 2022

- Implemented and maintained Data Center Infrastructure Management (DCIM) projects on distributed devices, which resulted in a 20% increase in availability, ensuring environmental and safety standards were consistently met and the development of a power load predictor / detector with 82% accuracy.
- Created and deployed AI models for energy prediction, demand forecasting, system ranking, and architectural assessment using preprocessed sensor data and scalable ML pipelines (e.g., neural networks) across distributed systems.
- Led and mentored a cross-functional team of 32 employees and managed external vendors in R&D to monitor the company's infrastructure (power usage, frequency, current, temperature, vibration, and airflow), yielding an 18% downtime reduction and minimizing failure recovery time.
- Supervised 4 cross-disciplinary design teams on 3 Data Center (DC) construction projects with a budget of \$35 million.

Data Center Infrastructure Designer, Solutions Architect

2016 – 2019

- Developed and debugged an AI-powered energy cost prediction using reinforcement learning and data modeling.
- Enhanced data analytics to optimize DC resources—power, cooling (HVAC, air, and water), and space—through the development of redesigned infrastructure (electrical distribution, UPS, Batteries, Diesel Generators, HVAC systems), resulting in a 22% reduction in energy consumption.
- Managed DCIM projects, a safety and security (S&S) team for DC cybersecurity where 6 software platforms integrated.
- Contributed to 16 data centers' reliability by preparing a preventive maintenance plan for critical infrastructures, and analyzing global compliance results against design standards.

Data Center Electrical and Control Infrastructure Designer

2014 – 2016

- Maintained, and troubleshooted power, cooling, smart meters, BMS, grid integration, robotics, and PLC systems.
- Developed and implemented a statistical model achieving 73% accuracy in predicting the stored energy in battery banks.
- Developed a fuzzy controller combined with a genetic algorithm using sensor data fusion to optimize the control process of the cooling and energy management system, reducing temperature/humidity fluctuation by up to 38%.
- Provided daily operational support by reading blueprints and electrical/mechanical schematics to inspect, maintain, and repair critical data center systems.

Arya Heavy Machinery - Caterpillar, Tehran, Iran

Power-Electrical Specialist Engineer

2013 – 2014

- Supervisor of the customer training department of the company
- Technical support of the sales team for heavy industrial power generation, UPS, and diesel generator deployments.
- Performing market analysis on power-electrical system segment, diesel generators

Professional AI & Research Experience

Computer Science and Engineering Department, University of Nevada, Reno, NV

- Graduate Research Assistant2022 – Present
- Designed and fine-tuned Transformer-based vision and language models (e.g., ViT, LLaMA, GPT-2) using subspace reparameterization, reducing communication and storage by 35% compared to LoRA, enabling on-device deployment.
 - Built and optimized dynamic ML systems using federated learning and split learning for heterogeneous clients, cutting training time by 80% in computer vision applications (Published in IoT journal).
 - Developed and deployed decentralized ML using dynamic tiering, reducing slower-side compute by 26% for deep learning models; with convex / non-convex convergence analysis (Presented in ICDCS conference).
 - Enhanced federated learning using distance correlation and patch shuffling in SplitFed (Published in the IoT Journal).
 - Engineered GNN-based predictive models for open source COVID data analysis, achieving 71% accuracy on 181M-node interaction graphs with optimized multiprocessing and graph-based data structures.

Connected Future Labs, Reno, NV

- Research and Development Scientist2024 – present
- Engineered automated multi-modal pipelines to ingest, clean, and harmonize high-frequency streaming sensor telemetry (including temperature, impedance, and inertial data) from distributed monitoring devices.
 - Extracted physiological biomarkers, e.g., HRV, sympathetic/parasympathetic responses, and neural features, analyzing biometric correlations to map signatures differentiating Mild Cognitive Impairment (MCI) from healthy cohorts.
 - Architected AWS S3, SageMaker cloud pipelines to autonomously audit multi-sensor data collection quality, engineer validation routines for signal anomalies, and drive end-to-end pipelines optimized for ML classification and prediction.
 - Led research and development (R&D) of AI-driven methods for monitoring and prediction of early-stage cognitive decline (Alzheimer) using wearable data, advancing early detection capabilities in healthcare applications.
 - Researched and analyzed state-of-the-art methods for the detection of Alzheimer's and its measured impact on the human body.

Battle Born AI, Reno, NV

- Machine Learning Scientist2025 – 2025
- Researched and analyzed LLMs to explore and implement clinical decision-support and business-focused applications.

Education

University of Nevada, Reno	2022 – 2025
Ph.D. in Computer Science and Engineering	
Dissertation: Distributed Machine Learning in Heterogeneous Edge Networks	
University of Tehran, Iran	2011 - 2014
M.Sc. in Electrical Engineering-Control Systems	
Thesis: Simultaneous State Estimation and Reinforcement Learning in Stochastic Matrix Games	
Shiraz University of Technology, Iran	2007 - 2011
B.Sc. in Electrical Engineering-Control Systems	

Research Experience and Publications

- Publications:
- A survey on self-supervised learning: Recent advances and open problems, Neurocomputing, 2025. (IF: 6.5)
 - Communication-Efficient Training Workload Balancing for Decentralized Multi-Agent Learning, IEEE ICDCS, 2024.
 - Advancing Electric Vehicle Ecosystems: A Survey of GenAI and Distributed ML Applications, GEI, 2025.
 - A Survey of Large Language Models: Evolution, Architectures, and Applications, Electronics, 14(18), 2025.
 - Speed Up Federated Learning in Heterogeneous Environments: A Dynamic Tiering Approach, IEEE IoT Journal, 2024.
 - Identification and Classification of Rheumatoid Arthritis Using AI/ML, Academic Press, 2025.
 - GenAI for Distributed Learning to Enhance Smart Grid Communication, Int. Journal of Intelligent Networks, 2024.
 - AI's Role in Diagnosis, Treatment, and Patient Care, CRC Press, 2024.
 - Feature collusion attack on PMU data-driven event classification, IEEE ISGT, 2024.
 - Federated learning based event classification for PMU data, IEEE PES General Meeting, 2023.
 - Simultaneous Learning and State Estimation in Repeated Matrix Games, ICITMCC, 2014.
 - SOLAR: Subspace-Oriented Lightweight Adapter Reparameterization for Scalable PEFT Compression, [Under Review]
 - A Survey on Hallucination in Large Language Models: Definitions, Detection, and Mitigation 2026 [Under Review]

Teaching Experience

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| • Teaching Assistant, University of Nevada Reno, NV | 2023 - present |
| Assisted in teaching core undergraduate & graduate courses Introduction to Big Data, Introduction to Data Mining. | |
| • Sematec, Iran | 2017 – 2022 |
| CDCP (Certified Data Centre Professional) & CDCS (Certified Data Centre Specialist) | |
| • Sematec, Iran | 2018 – 2021 |
| Data Center Standards & Best Practices (TIA-942, BICSI 002-2019, BICSI 009-2019) | |
| • Sematec, Iran | 2019 – 2020 |
| CDCE (Certified Data Centre Expert), | |
| • Teaching Assistant, University of Tehran, Iran | 2011 – 2014 |
| Modern Control | |

Professional Certifications, Awards and Honors

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- GSA Outstanding Graduate Student Advocacy Award, UNR, 2025
 - GSA Outstanding Graduate Researcher Award, UNR, 2024
 - Awarded the TechWise program scholarship by TalentSprint Inc., supported by Google (2023 - 2024)
 - Python for Data Science and AI by IBM (Coursera 2020)
 - Deep Learning with Python and PyTorch by IBM and Deep Learning with TensorFlow by IBM (edX 2020)
 - Certified Data Centre Specialist, Professional & Expert (CDCS, CDCP, & CDCE), epi, Iran & Malaysia 2016-2017
 - DC102 (Applied Data Center Design and Best Practices | BICSI), TUV Academy, 2016
 - Student of National Organization for Development of Exceptional Talents (2000-2007)
 - ITIL, Linux, EN50600 standard, Cooling Infrastructure Designing, CDCD (Certified Data Center Designer)

Professional Activities and Service

Graduate Research Assistant – NSF REU Site: Cross-disciplinary Research Experience for Undergraduates on Big Data Analytics in Smart Cities, University of Nevada, Reno 2022–2023

- Contributed to an NSF-funded initiative (Award #1950485) by developing self-supervised machine learning algorithms for classifying power system events using real-world Phasor Measurement Unit (PMU) data.
- Addressed challenges associated with noisy, incomplete measurements and validated models for use in smart grid applications.

Peer Reviewer

2023–Present

Regular reviewer for leading journals and conferences, including:

- IEEE Internet of Things Journal
- IEEE Journal of the Communications Society
- IEEE Transactions on Wireless Communications
- IEEE/ACM Transactions on Networking
- IEEE Transactions on Network Science and Engineering
- Nature Communications
- ACM SIGMOD Conference
- Journal of Biomedical Informatics

Skills & Core Competencies

Key Skills: Data Center Operations & Infrastructure • Machine Learning & AI • Mathematical Analysis & Optimization

- Facilities & Infrastructure: Electrical Distribution, UPS, Diesel Generators, HVAC (Cooling), BMS/PLC, DCIM, Grid Integration, Vendor Management, Outage & Emergency Response, Blueprint, Facility Compliance, AutoCAD.
- Machine Learning & AI: Supervised & Unsupervised Learning, Reinforcement and Distributed Learning, Regression, Tree-based Models (XGBoost), Large Language Models (LLMs), GenAI, Parameter-Efficient Fine-Tuning (PEFT).
- Mathematical Analysis & Optimization: Predictive Modeling, Optimization & Statistics, Data Structures, Sorting & Searching Algorithms, Graph Algorithms (DFS, BFS), Algorithm Development.
- Programming, Tools & Frameworks: Python, MATLAB, Simulink, PyTorch, TensorFlow, NumPy, Pandas, Flower, Git, LaTeX, Linux, Computer Networks.